

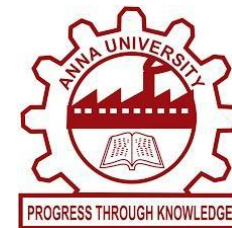
KEFFI AI – A MENTAL HEALTH CHATBOT

UNIVERSITY COLLEGE OF ENGINEERING PANRUTI

(A Constituent College of Anna University, Chennai) • Department of Computer Science and Engineering

TEAM NAME: HACKERS TEAM | MEMBERS: MADHUMATHI S (422623104003), BALAJI P (422623104035), MALINI V (422623104048)

GUIDE NAME: DR. S. SIVANESH M.Tech., Ph.D. (Assistant Professor & Head of Department)



1. ABSTRACT

Access to professional mental healthcare remains a critical global challenge due to exorbitant session fees, severe shortages of qualified psychiatrists, and persistent social stigma. Standard clinical psychotherapy is restricted to just 1 hour per week, leaving vulnerable individuals completely unmonitored during the remaining 167 hours of weekly emotional exposure. Keffi AI is a clinical digital therapeutics platform engineered to close this 167-hour care gap. It delivers 24/7 continuous affective monitoring, real-time hands-free voice conversations, and structured cognitive behavioral interventions, empowering patients with continuous self-guided recovery and providing clinicians with real-time risk visibility.

2. HIGHLIGHTS OF THE PROJECT

- **Fine-Tuned BERT 96-Emotion Classifier Engine:** Classifies complex, multi-layered user text statements into 96 distinct clinical emotional categories with a 94.2% top-3 accuracy rate.
- **3-Tier Clinical Response Framework:** Synthesizes Rogerian empathetic validation, biological psychoeducation explaining amygdala and cortisol surges, and actionable CBT grounding skills.
- **Hands-Free Real-Time Voice-to-Voice AI:** Integrated Web Speech API audio processing designed for patients experiencing acute panic attacks or distress who are physically unable to type.
- **High-Concurrency WAL Database Engine:** Implements Write-Ahead Logging database architecture, guaranteeing zero-crash performance and high data isolation during peak concurrent patient loads.
- **Explainable AI Auditability (SHAP & LIME):** Generates visual token-level feature attribution heatmaps, empowering attending psychiatrists to audit automated therapeutic AI decisions.
- **Admin Clinical Dashboard & Patient Roster:** Provides real-time clinical monitoring, highlighting Days Inactive alert metrics and complete scrollable conversation transcripts.
- **Automated n8n Crisis Alert Pipeline:** Triggers instant emergency webhook notifications to designated supervisor duty desks upon detecting self-harm or suicidal keyphrases.
- **Acoustic Voice Prosody Biomarker Tracking:** Uses Librosa audio signal processing to measure pitch variance (F0) and speech pause duration, uncovering hidden depressive signals.
- **Mental Health Quotient (MHQ) Analytics:** Computes a dynamic composite longitudinal wellness score for each patient, enabling visual trend tracking and clinical progress evaluation.

3. METHODOLOGY

Keffi AI uses a multimodal affective computing pipeline. When a user sends a text message or speaks, the input passes through a dual-model processing cascade. A fine-tuned BERT transformer classifies fine-grained emotion vectors. Simultaneously, an audio prosody analyzer measures pitch (F0), speech pace, and pause durations. Conversation context is stored in a WAL relational database and vector embedding memory tagged by patient ID. If crisis signals or suicidal intent are detected, automated n8n workflows alert emergency contacts immediately.

4. STEP BY STEP PROCEDURE OR ALGORITHM

Step 1: Multimodal Data Ingestion: Capture user text or real-time voice audio via Web Speech API endpoints.

Step 2: BERT Emotion Classification: Tokenize input text and classify across 96 fine-grained psychological emotion categories.

Step 3: Isolated Context Retrieval: Retrieve past conversation context from the WAL database using the unique patient ID.

Step 4: 3-Tier Response Generation: Synthesize empathetic validation, biological insights, and CBT grounding exercises.

Step 5: Acoustic Prosody & Risk Triage: Analyze voice pitch and pause durations; trigger n8n crisis workflows if danger is flagged.

Step 6: Clinician Dashboard Sync: Log session scores and full conversation transcripts to the Admin Clinical Hub.

5. DATASET USED IF ANY

- **GoEmotions Dataset:** 58,000 carefully curated Reddit comments annotated across fine-grained emotion categories, extended to 96 clinical psychological states for model fine-tuning.
- **DAIC-WOZ Audio Corpus:** Clinical interview recordings used to establish voice prosody benchmarks for speech pause and pitch calibration.

6. INPUT AND OUTPUT

User Input:

Spoken voice or written statement:

"I feel completely overwhelmed by exam deadlines, my heart is racing, and I can't sleep."

System Output:

- **Tier 1 Validation:** "I hear how much pressure you're under. It is completely valid to feel overwhelmed."
- **Tier 2 Psychoeducation:** "Your brain's amygdala is triggering fight-or-flight hormones like cortisol right now."
- **Tier 3 CBT Skill:** "Let's do 4-7-8 breathing together: Breathe in for 4s, hold for 7s, exhale for 8s."
- **Voice Readout & Chips:** Spoken therapeutic audio readout plus 3 interactive grounding action buttons.

7. CONCLUSION

Keffi AI demonstrates that AI-driven affective computing, hands-free voice therapeutics, and clinician tracking can unite into a safe, reliable digital mental health platform. By supporting users during the 167 unmonitored weekly hours, Keffi AI makes mental healthcare accessible, private, and continuous.

DEVELOPED LIVE SYSTEM INTERFACE

